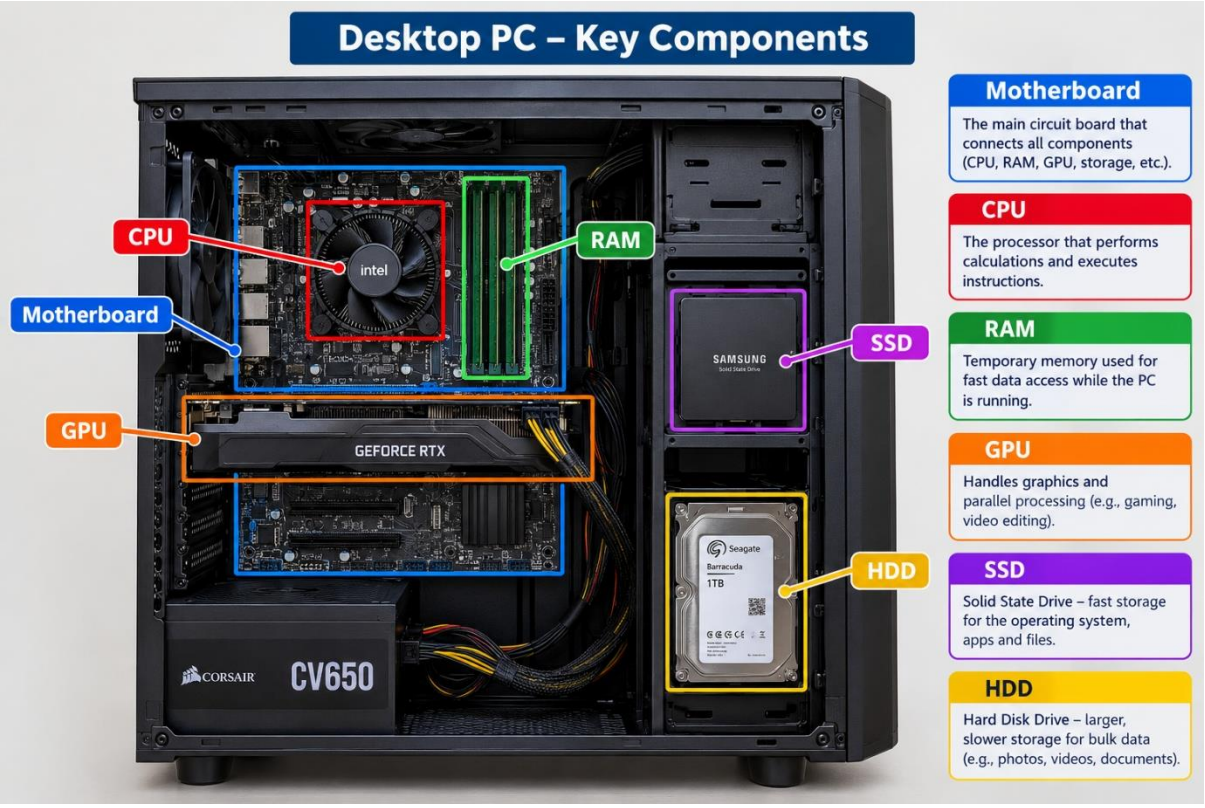
Day-1

Overview of Hardware Components Theory.

- Introduction to motherboards , CPUs , Ram , GPUs , storage (HDD,SSD).
- Importance of each component in a computer system .
-Assignment :- Create a diagram of a PC with labeled hardware components .

Q **-1 :-** Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

Ans-A desktop PC is made up of several important components that work together to run programs and games. The motherboard connects all the components, the CPU processes instructions, the RAM temporarily stores data for quick access, and the GPU handles graphics and images. The HDD provides large-capacity storage at a lower cost, while the SSD offers faster storage for quick boot times and faster game loading.



Q-2 :- Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).

Ans -SSDs are much faster and more durable than HDDs, while HDDs remain useful for large, low-cost storage in desktops and backup drives.

Below is a clear comparison table followed by deeper explanations to help you understand when to choose each type.

Comparison Table: HDD vs SSD

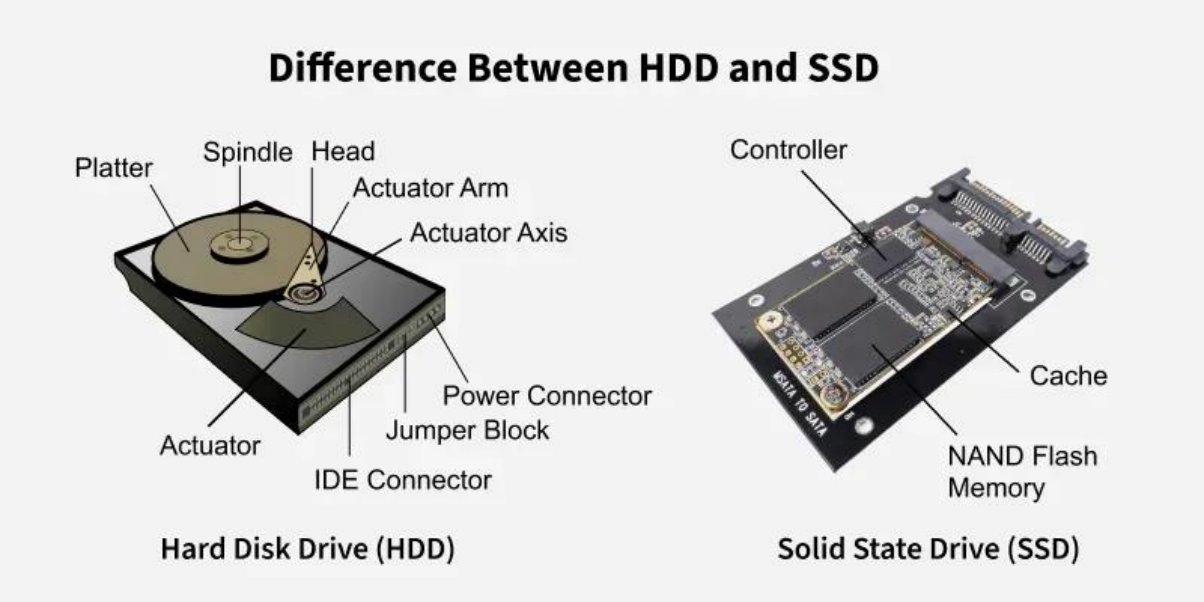
HDD vs SSD – Comparison

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Image		
Speed	• Slower • Typical read/write: ~80–160 MB/s	• **Much faster** • Typical read/write: ~500 MB/s to 7,000+ MB/s (NVMe)
Durability (Shock Resistance)	• More fragile • Has moving mechanical parts (spinning disks), so more vulnerable to shocks and drops	• **More durable** • No moving parts (uses flash memory), so highly shock-resistant
Typical Use Cases	• Best for large, low-cost storage • Commonly used for bulk data (movies, backups, archives) • Often found in desktop PCs, external drives, and budget systems	• Ideal for fast performance (quick boot, faster app/game loads) • Commonly found in modern laptops, gaming PCs, and high-performance desktops
Other Notes	• More storage for the same price (e.g., 1–4 TB is affordable)	• More expensive per GB (e.g., 512 GB – 2 TB, though prices are decreasing)

HDD		SSD
	vs	

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Slow speed	Fast speed
Cheaper	Expensive
Less durable	More durable
Use more energy	Use less energy
20TB max size	100TB max size



How They Work (Why the Differences Exist)

- **HDDs** store data on spinning magnetic disks. A mechanical arm moves to read/write data, which limits speed and makes them vulnerable to physical shock.
- **SSDs** store data in NAND flash chips. With no moving parts, they access data almost instantly and resist physical damage.

Which One Should You Choose?

- Choose **SSD** if you want:
    - Fast boot times
    - Quick app/game loading
    - Better durability
    - Silent operation
    - Improved laptop battery life
- Choose **HDD** if you need:
    - Large, affordable storage (e.g., 2–10 TB)
    - Backup drives
    - Media libraries (movies, photos, archives)
    - Budget-friendly desktop builds

Practical Examples (India-relevant)

- **Laptops sold in India** increasingly use SSDs because they make even budget systems feel fast.
- **Gaming PCs** almost always use an SSD for the OS and games, sometimes paired with an HDD for extra storage.
- **Office desktops** may still use HDDs for cost savings when speed is not critical.

If you want, I can also help you choose the right storage type for your specific device or use case.

Q-3 :- Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC

Ans -

1. CPU (Central Processing Unit)

The CPU acts as the brain of the computer, processing game instructions and managing tasks such as player movements, enemy actions, and game logic. A powerful CPU helps the game run smoothly and respond quickly to player inputs.

2. RAM (Random Access Memory)

RAM temporarily stores the game data that the CPU needs to access quickly, such as maps, player information, and other game resources. More RAM helps the game run smoothly and reduces lag when multitasking.

3. GPU (Graphics Processing Unit)

The GPU renders the game's graphics, including characters, weapons, buildings, shadows, and visual effects. A powerful GPU provides better graphics quality and smoother gameplay with higher frame rates in games like PUBG or Free Fire.

Q-4 :- Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

Ans - To upgrade my PC to run the latest version of FIFA (EA SPORTS FC) smoothly, I would consider upgrading the following hardware components:

1. GPU (Graphics Processing Unit): I would upgrade to a powerful graphics card to improve graphics quality, reduce lag, and maintain smooth gameplay.

2. CPU (Central Processing Unit): I would choose a faster processor to handle game calculations, player movements, and other tasks efficiently.

3. RAM (Random Access Memory): I would upgrade to at least 16 GB of RAM for smooth gameplay and better multitasking.

4. SSD (Solid-State Drive): I would install an SSD to reduce game loading times and make the PC start faster.

5. Cooling System: I would improve the CPU cooling system to prevent overheating and maintain performance during long gaming sessions.

6. Power Supply Unit (PSU): I would upgrade the power supply if necessary to provide enough power for the new graphics card and other components.

Conclusion: I would prioritize upgrading the GPU, CPU, and RAM because they have the greatest impact on gaming performance. I would also ensure the PC meets the game's system requirements. For example, the latest EA SPORTS FC 27 lists a GTX 1660 or RX 5600 XT and 12 GB of RAM as recommended specifications.